

Chalk's Resistance

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This document has been translated by me, trying to get a document like the original without adding or subtracting anything. Please read the Revision and further improvement to see my evolution and present judgment of this work.

1.Introduction.

The aim of this coursework is the approximate measurement of the different stresses normal chalk can withstand. For the proper measure of the properties, it was necessary the design and manufacturing of the test apparatus, whose details are explained in each section. My previous research showed that there was not any information available concerning the mechanical properties of chalk.

1.1 Objectives

The objective of this work is not the accurate measurement of the different maximum stress, to do so, it will involve the use of a set of equipment (such as lathes, hydraulic presses...) with which I could create more precise apparatus and obtain more accurate measurements.

Thus, my main objective for this work is to be able to solve the different technical problems that appeared during the design, manufacturing and experiment of the different apparatus that allow me to obtain the most accurate measurement of the maximum stresses of the chalk.

1.2 Working Method

I have used the same working method for every different way of loading:

First, it is necessary the visualization and design on how the type of loading is going to be measured (depending on which stress) in the most accurate way and trying to minimize the influence of external factors: friction forces, the appearance of other types of stress different to the one we want to measure...

Once I have decided which is the most efficient and effective way to take the measurement, I started to manufacture it.

Then, when the test apparatus is finished, we start the experiment of the chalk itself:

If the stress is measured properly, i.e. without the presence of any other stress, I would proceed to the measurement of 3 to 4 measurement of the fail stress, except for the torsion, when I used 5 different measurements. Once I have all the measurements I thought about the accuracy of them and if they agreed with my experience, if they were completely unreal, I would come back again to the first step, redesigning or doing the experiment again, this time with the wisdom of the failure.

If in the first try of the test I realized that the apparatus was faulty and did not help me to measure the actual stress (other stress showed up) I studied the reasons for that to happen and I redesigned my apparatus trying to avoid the spotted errors.

2. Bending

This was probably the easier stress to measure because it did not require the manufacture of great complexity apparatus, as you could see in the following pages.

2.1 Framework

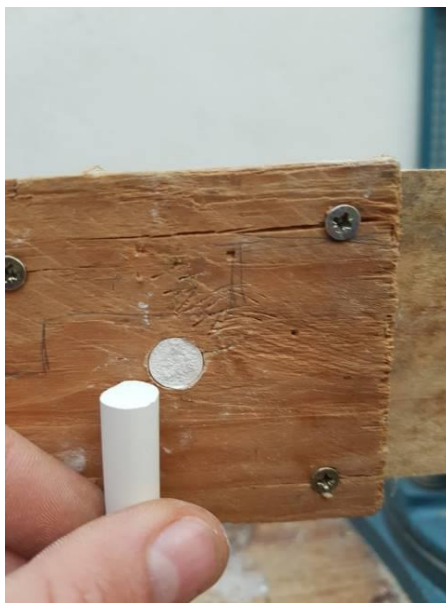
For measuring this stress, I used for the manufacturing of the test device a wood table hold by vice to the workbench, orange nylon cord, a 1.5l water bottle and, of course, chalk. The display can be seen in the photo below:



I just simply use the wood plate with a 1 cm hole ($\varnothing$ of chalk). Afterwards, I insert the chalk in the hole. From a 53mm distance (left picture) I attached the bottle, which was filling up until the chalk broke. Then, I use a kitchen scale (around ± 2 grams of precision) to weight the bottle, obtaining the kilograms of force [kg-f] that the chalk can withstand.



The left photo was taken in one of the tests, you can see in the scale the weight of the bottle was 538 grams, which means the chalk requires a 0.538 kg-f hanging from 53mm distance to break.



You can see in the left figure the characteristic breaking way obtained with bending stress; the breaking is almost perpendicular to the chalk surface.

2.2 Results.

In this particular case I make the experiment 4 times, obtaining the following results:

Test's number	Failure force	Failure torque
1	0.538 kg-f = 5.2724N	0.2794N·m
2	0.491 kg-f = 4.8118N	0.2550N·m
3	0.497 kg-f = 4.8706N	0.2581N·m
4	0.476 kg-f = 4.6648N	0.2472N·m

2.3 Conclusion.

From the previous table, we can deduce that the force required to break the chalk in bending is around 0.5 kg-f (4.9N), although there is a $\pm 0.04\text{kg-f}$ uncertainty, this means that the mean moment of the fore was 0.26 Nm, which is a relatively low value, agreeing with its brittle behaviour and confirming why it is so easy to break them with the hands when we apply a bending moment stress.

Another thing to notice is the small difference between the maximum strain (0.538kg-f), the mean (0.5kg-f) and the minimum strain (0.476kg-f), this shows us this measurement is quite reliable. The reason is that the loading way is really easy to measure, in the following experiments we will see this error margin (variance) is getting bigger, mainly due to the use of more sophisticated apparatus which implies that the tests are more subjected to more external factors, thus losing precision.

3. Traction.

For the measurement of this type of loading was not require a huge amount of different materials neither, though the repeatability of the test was more complex than the previous one, as I will explain. it was required to hold the chalk use tow double knots. In general, I had no much trouble to measure it.

3.1 Framework

In this case, I use a 1 mm (approx.) nylon cord for the measurement of the force, strong enough to hold the weight of the water bottle which I used to measure the fail force. The cord was tied to the chalk using two clove hitch knots (constrictor knot).



As can be seen in this picture, I needed to make two knots in each end, i.e. first I made a clove hitch, and instead of hanging the chalk directly from it (this caused the misaligned of the chalk when hanging creating an angle with respect to the vertical axis), I used the other end of the cord to make another clove hitch, thus both ends are opposite to each other creating a loop in the middle from where I attached the weight and hold it in a vertical position.



In the left image, it can be seen that the chalk was attached to a fixed wire from the top, and in the bottom end, there is a hook from where the weight was hung.

It can be seen the result of having tied tow knots, the force is just vertical, avoiding the misaligned of the chalk and making the force vector completely parallel to the chalk axis.



The next picture shows the full set up of the experiment, the chalk hanging from one end and in the other the hanging weight (water bottle) which was being filled up gradually until the chalk's breaking. Then, the bottle was weight using a kitchen scale.

The grey rope which appears on the left of the image is used to avoid the free-falling of the water bottle as this could make the bottle splashing or spilling and make the test fail



In the picture above, one can appreciate the characteristic breaking surface of the traction load, which differentiates because both fracture surfaces are almost parallel.

3.2 Results.

The results obtained in the experimental measurements are:

Test number	Failure force
1	3.005kg-f = 29.449N
2	3.061kg-f = 29.997N
3	2.770kg-f = 27.146N
4	2.818kg-f = 27.616N

3.3 Conclusions.

With these experimental data, I can conclude the chalks requires around 2.913kg-f (28.552N) for its breaking in traction, with a precision of around ± 0.200 kg-f, which is a relatively acceptable value for a completely homemade experiment and with just a few samples.

It is also worth mentioning that this measurement agrees with my intuitive measurement while trying to break chalk using traction. Though such force can be overestimated since when you are loading the chalk in traction, the fingers also compress it (to avoid its slipping) which make the “feeling” force required to break if to appear bigger than 2.9kg-f.

4. Pure shear.

In this case, I start to have some trouble while measurement the stresses because it was required the manufacturing of more complex apparatus to take the measures.

I make a first apparatus that can be seen in the next image:

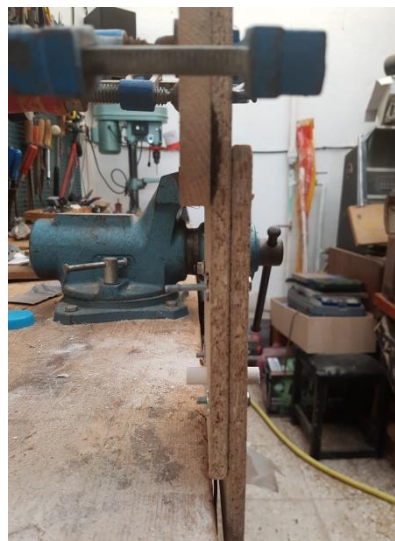
As can be seen in the picture, the support is made from two parts, one fixed with clamps (back) and the other free to move (front) which is attached to the fixed using two rails while making a concentric 1 cm $\varnothing$ hole from where the chalk is passed through to measure its strength.

The problem with this system was that: hanging the weight from the moving part generated a certain moment (in the horizontal axis) big enough to not enable a proper measurement. This was because the tables had a small contact surface and when the moving part rotated (due to the moment) it got stuck with the rails, increasing considerably the friction force. On the other hand, if I made the rails looser, I reduced the friction force, but then the moving table could separate from the fixed one, creating a bending load. I solve all these problems in the following way:



4.1 Framework.

For avoiding the previous error I used a table with bigger surface area and instead of using two rods, it was held by 4, which minimize the moment (in the case I was right and that was the trouble source)



In the two pictures above, one can see perfectly the apparatus used in the test. The main difference with respect the first device is that the contact surface of the table is bigger; this allowed me to use 4 guides instead of two, which also improved the device.

Once the test device was manufactured, the test was straightforward, since I only have to introduce the chalk in the concentric hole and hang from the bottom of the sliding part a water bottle, which was then weighted. The biggest drawback, in this case, is that for weighting the water bottle I just used a normal scale, which gives me an awful precision of around ± 0.1 kg, even though, this was enough to give me an approximate idea of the strain the chalk was capable of withstanding in this type of loading.



In this image, one can see in detail the kind of breaking the pure shear produce in the material, in which both faces of the fracture rub each other.

4.2 Results

The experimental results are:

Test number	Failure force
1	5.8kg-f = 56.84N
2	5.6kg-f = 54.88N
3	5.6kg-f = 54.88N

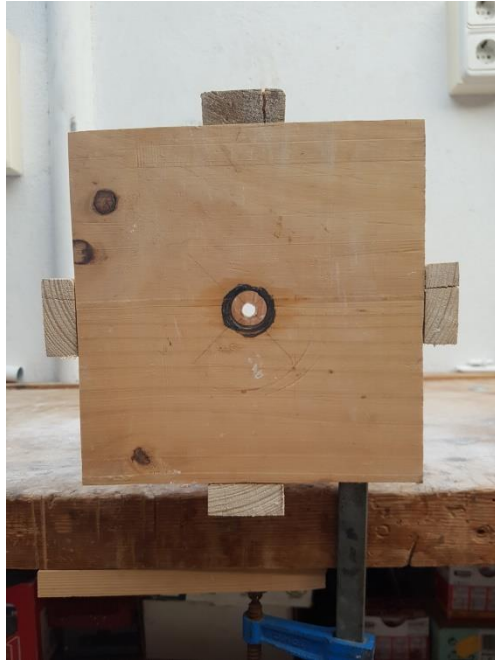
4.3 Conclusion.

Despite the low precision the scale offered, the data was consistent, the required force to break the chalk in pure shear is round 5.56kg-f (55.53N). This value agrees with the intuitive value one can estimate while breaking the chalk with the hand. Although, as I clarified in the introduction, these tests only give us a broad estimation of the real value.

5. Torsion.

This had been one of the most complex stresses to measure since to achieve a torsion breaking instead of bending, I had to design a machine that reduces to a minimum the bending stress. The first two prototypes only allowed me to get some experience in obtaining the measurement since they did not give me real measurement (too much friction) although minimizing the bending. Finally, I manufacture an apparatus that give me reasonable values (more or less).

5.1 Framework.



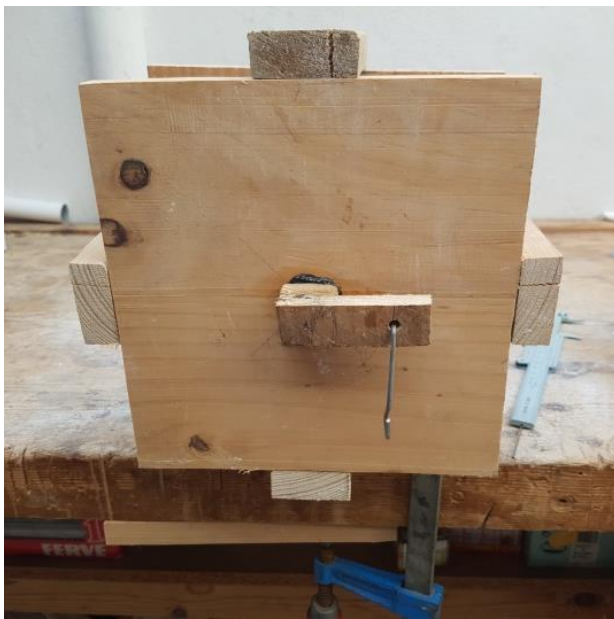
To avoid the bending, I used a kind of homemade bearing; this was placed in the centre of a table which was attached to a second one using 4 pieces of wood, there was a 1 cm hole at the end of the bearing-rod, as well as in the other table, such as I could place each end of the chalk in one hole and glueing it together, fixing it.



In this photo, one can see the homemade bearing-rod. To minimize the friction I spread some grease on the rod.



One can also see that the distance from the hanging weight (water bottle) to the centre of the rod was 4cm.



On the left picture, you can see the front view of the device. The weight was hanging down from the hanging steel wire.



On the right image, we can see the final position of the chalk once it had been glued to the table in one end and to the bearing-rod in another. That picture was taken after breaking the chalk. To get as many reliable values as possible (because the bearing system was inefficient) first I measure the bearing force, 0.05kg-f, and then I take this out from the obtained value.



In these two images, it can be seen the type of fracture (helicoidally) that this type of stress creates. One can also appreciate how the chalk is glued to the bearing-rod.

Results.

The results obtained experimentally are:

Test's number	Failure force	Failure torque:
1	0.285 kg-f = 2.794N	0.011N·m
2	0.258 kg-f = 2.528N	0.010N·m
3	0.350 kg-f = 3.430N	0.013N·m

5.2 Conclusion.

From the results, we can say that the force required to break a chalk bar through torsion is around 0.297kg-f (2.917N), which means the required torque to break it is around 0.116 Nm. So, how is it possible that when we broke it with our hands the force “feels” more than 0.3kg-f?

The reason for that is when we try to break with our hands in torsion; the distance from which we apply the force is very small, half of the diameter of the chalk (0.5cm), so to break in this way we need around 2.38 kg-f (23.339N) to create the same torque.

6. Compression.

This has been, without doubt, the hardest stress to measure technically, due to different reasons:

- The first reason is that the chalk can withstand a great load in this type of loading (according to the measurements around 25.5 kg-f), so if we would like to take the test using multiple samples and trying to balancing 3 or 4 chucks it would require more than 100 kg to break them.
- The second reason is that was necessary to have the chalk completely vertical when applying the force, which was tricky to achieve as the cross-section area was very small compared with the height.
- The third one is the lack of precision since I was using free weights because the required force was so big that I had no other possibility to measure it.

First I tried to use a vertical rail-guided system, but for lack of precision, this system was a total failure. So inspired by the hydraulic presses, I had the idea of used a similar system based on a cylinder-piston operation:

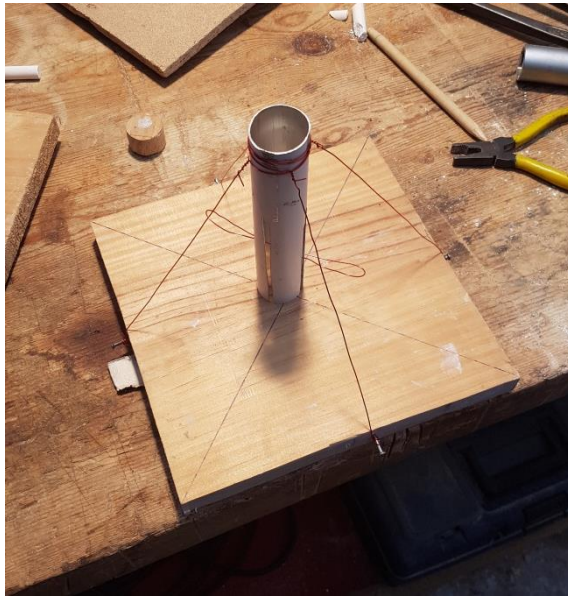
6.1 Framework.

The first thing I did was to make sure that the surface where I was going to build the apparatus (workbench) was completely flat:

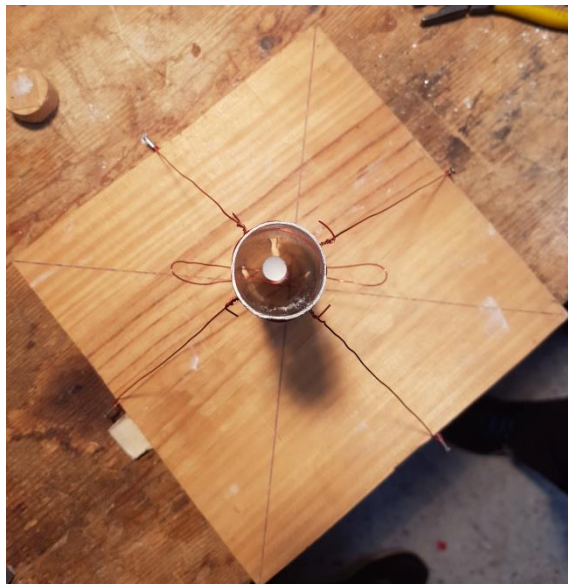


Once I had checked that the table was completely flat, I started to build my apparatus:

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The cylinder was simply put on a table. To avoid its overturning I attached 4 wires to it as guides. The cylinder also had 4 gaps so I could observe the chalk while testing and it helped me to put and maintain the chalk in its place.



In this picture one can see how the system is held, with the 4 wires that maintain the cylinder vertical and the chalk, which had attached a wire (loops) to it. The wires were inserted in the gaps of the cylinder to put the sample completely vertical.



In this figure can be seen the box spanner used as a piston. To get a better fit I roll up a couple of turns of adhesive tape. Between the piston and the chalk, I placed a wood cylinder for uniform pressure distribution.



Here you can see the complete system. In the background, there are the free weights I used to measure the force. This is why this is possibly the experiment in which less precision I have used as well as for being the more complex.

Once the whole set up was ready, I placed the free weight on top of the piston, for which I was helped by my father since I was balancing the free weights to avoid overturning while he was placing the free weights one by one.



In these figures, one can see the type of characteristic breaking, parallel to the cylinder axis, as well as the wire-guided system (left and right photos) and the situation of the cylinder-piston once the chalk was broken (centre)

6.2 Results.

The results obtained experimentally are:

Test's number	Failure force
1	26.0kg-f = 254.8N
2	24.0kg-f = 235.2N
3	26.5kg-f = 259.7N

6.3 Conclusion.

Although the free weights reduce significantly the accuracy, from the obtained values we can conclude that the value for breaking the chalk is around 25.5kg-f (249.9N) which means that with only 3 chucks would be possible to hold a 70kg man. I find it striking coming from this modest material, mostly used for educational purposes.

To the question: Would be possible to build a building from chalk? I would answer that theoretically, it is possible but if and only if all the chucks are load in compression, which is technically impossible.

7. General conclusions.

Though the obtained results only give us an idea of the resistance of the chinks, I will remark what this work has meant to me in a problem-solving way, facing the problems that were appearing and trying to solve them in the more simple, efficient and technically possible way (with my limited resources).

Said this, the results seem coherent with the reality but some of them surprised me, especially the compression and torsion ones.

8. Revision and further improvements

One of the first thing that came to my mind while translating and revising this work was the context in which it was made. This was one of my coursework in the first year of my degree in Spain and one can see a lot of flaws and mistakes:

Starting for the presentation, which is relatively bad, none of the figures has headings and are presented in a “peculiar” way. The explanations are not really impressive neither. The main reason for this is that the work was tailored to the professor that proposed the coursework, which had a strong intuition for engineering and preferred a concise explanation with a lot of pictures. Apart from this, the devices are relatively simple and only a few images are enough to describe them, though my point with it was to be able to show the thinking process I were through, showing my possibilities and what I would learn later: the design process.

Another topic to discuss is the lack of scientific rigour with the measurements and the way I handled them. I also had some misconceptions between the ways of loading and did not fully understand then, as for example, if you have bending inevitably you will also have shear. Also, the methods of testing are not the most proper for this kind of material, as a ceramic one, it requires different treatment such as flexural strengths test instead of tension. But we cannot forget that all of this was completely unknown at that time, I was just starting university and most of this material was not been given to me.

So, why do I take my time and effort to translate this document? Although imperfect and with a lot of flaws it still shows one of my biggest virtues: **the ability to, for a given task, be able to arrive at the best product or conclusion with the given resources.** One can see my mental process reading this document, realising how I solve some of the problems in a really simple way, one can see how I track the problem and fell my intuition while solving the issues that were appearing in my way and how the immature (in that time) version of me was able to arrive at concrete and correct analysis more for intuition than knowledge.

Now, I am a more mature engineer, more aware of all the challenges that one has to overcome in order to create a great product and with a lot of more knowledge but that engineering intuition, that spark that ignites inside me while engineering, that remains intact.